

A Unified Deep Learning Approach for the Segmentation of Breast Masses and Calcifications

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Abstract—Breast cancer is a leading cause of mortality among women and early detection using computer aided diagnosis (CAD) systems assist in improving the survival rates. Segmentation is a crucial step in CAD systems as it helps to accurately extract breast abnormalities. This study proposes a unified deep learning based approach that can accurately segment breast masses and calcifications in contrast to studies that propose approaches to segment only a single type of abnormality. The proposed framework uses a modified Hybrid Transformer U-Net (HTU-Net) for segmentation of masses and U-Net for segmentation of calcifications. Quantitative results show that the U-Net achieved a Dice Similarity Coefficient (DSC) of 0.8022 and a precision of 0.8958 for segmentation of breast calcifications. For segmentation of breast masses, the modified HTU-Net achieved a DSC of 0.6064 and a precision of 0.5988. These results demonstrate the effectiveness of the proposed unified approach in segmentation of breast masses and calcifications, highlighting its potential in being integrated to CAD systems.

Keywords—breast cancer, mammograms, masses, calcifications, segmentation, U-Net

I. INTRODUCTION

Breast cancer is one of the main causes of mortality among women worldwide with about 2.3 million women diagnosed with breast cancer and 670,000 deaths globally in the year 2022 [1]. The five-year survival rate for localized breast cancer is about 99% [2] and it proves that early detection of breast cancer improves survival rates significantly. Diagnosis of breast cancer is done through different imaging techniques such as mammograms, ultrasound and magnetic resonance imaging (MRI) [3].

Computer Aided Diagnosis (CAD) systems assist radiologists in detecting breast cancer and classifying into different stages [4]. Segmentation is an important step in CAD systems as it is used to accurately extract the region of interest (ROI) from mammograms [5]. These ROIs often include two types of abnormalities namely, masses and calcifications [6]. Several studies have been carried out for segmentation of masses [7] [8] [9] and segmentation of calcifications [10] [11] [12] separately. However, there exists a gap in research which explore

the possibility of segmentation of both masses and calcifications through a unified approach.

This study proposes a novel segmentation method to accurately extract the ROIs of mammograms which are useful for diagnosis of breast cancer through a deep learning (DL) based approach.

II. LITERATURE REVIEW

Segmentation of ROIs for breast cancer diagnosis has been explored by numerous studies. These studies have proposed different approaches such as traditional image processing methods, machine learning (ML) based methods and DL based methods for segmentation of masses and calcifications.

Dominguez et al. [13] proposed two segmentation methods based on dynamic programming and constrained region growing to obtain mass contours. The obtained contours were used for feature extraction and were then used as input to classifiers for prediction of malignancy of masses. Iterative pixel based classification methods such as Markov random field, Gibbs random field, region growing and region clustering are also used for mass segmentation from mammograms [14]. A fuzzy thresholding based segmentation method was proposed for segmentation of breast masses by Sameti et al. [15]. Dong et al. [7] proposed a segmentation method for breast masses using an improved vector field convolution snake and was followed by a feature extraction. These features were used as input to several classification techniques such as support vector machines (SVM), genetic algorithm SVM and particle swarm optimization SVM.

Methods such as multi resolution fuzzy c-means segmentation [16], difference of Gaussian filters and morphological filters [17] and local area thresholding [10] have been used for segmentation of calcifications. A fully automated pipeline is proposed in [18] to segment calcifications inside the breast region with a novel texture filter. However, the above methods encountered limitations due to factors such as the complexity of breast images with

high density tissues and irregularity of edges in breast masses [19].

Due to the limitations of above traditional image processing methods and ML based approaches, DL based approaches for mass and calcification segmentation were proposed. Wang et al. [20] suggested a multi-level nested pyramid network for dealing with the limitations in the inconsistency between classes in breast mass segmentation. The approach proposed by Pezeshki et al. [21] segmented and extracted the speculated regions with the mass in mammograms. U-Net models have been extensively used for mass and calcification segmentation and have achieved better performance comparatively [8] [9] [22]. Also, modified U-Net architectures such as recurrent attention U-Net [23], simple context U-Net [24] and modified fully convolutional network [12] have been proposed for segmentation of breast calcifications in mammograms.

However, there is a lack of studies that focus in segmenting both breast masses and calcifications through a unified approach. This study addresses this gap through the proposed framework which accurately segments both masses and calcifications using a DL based approach.

III. METHODOLOGY

The proposed segmentation methodology has the objective of accurately extracting the breast masses and calcifications in mammograms. The performance of the proposed approach was evaluated using the publicly available INBreast dataset [25] which provides high resolution mammograms in the Digital Imaging and Communications in Medicine (DICOM) format. The dataset also consists of the annotations for masses and calcifications in mammogram images.

A. Preprocessing

The mammogram images from the dataset, originally in the DICOM format were converted to Portable Network Graphics (PNG) format prior to processing. The dataset contained mammograms from both Cranio Caudal (CC) and Mediolateral Oblique (MLO) views. Initially, the pectoral muscle present in the MLO view images was removed by using the pectoral muscle annotation as this muscle can cause misdiagnosis [26].

The images of the dataset were of two different types. Therefore, they were converted to standardized images of size 1024 x 1024 using a few preprocessing steps as follows. The initial step was creating a rectangle that bounded the foreground area of the image which contained the breast by detecting the largest external contour in the image. Then the cropped mammogram was obtained by cropping the original mammogram according to the dimensions of the bounding rectangle obtained previously. Then a proportional resizing approach was followed by keeping the aspect ratio of the image a constant. Then a laterality based zero intensity padding was inserted to the images in order to acquire the standard target size. Through these steps, the standardized images were obtained without any distortion and by preserving the anatomical consistency of the breast.

Due to the structural differences of breast masses and calcifications, two different enhancement pipelines tailored for each type were used.

For enhancement of breast masses, Contrast Limited Adaptive Histogram Equalization (CLAHE) was used with a global threshold value to highlight the ROIs. The visualization was further enhanced by applying magma colour mapping and three images were obtained by separation of the three colour channels red (R), green (G) and blue (B). Finally, three fused images were obtained as red-green (R-G) fused, red-blue (R-B) fused and blue-green (B-G) fused images. According to the enhancement measures and similarity measures of the images obtained, the B-G fused was considered the most enhanced. Hence, the B-G fused image was used for segmentation of masses.

In order to enhance visualization of calcifications, a high-pass filtering technique was used by modifying a Gaussian low pass filter. Afterwards a rectified image was obtained and a secondary Gaussian smoothing was performed on the rectified image. An improved visualization of the suspicious calcifications was obtained by overlaying the obtained binary mask on the standardized mammogram. The resulting images with enhanced breast calcifications were used as inputs for the segmentation of calcifications.

Fig. 1 shows the comparison of original mammogram with the breast mass enhanced image and the breast calcification enhanced image.

B. Segmentation

The breast masses and calcifications have significant differences in their structure, distribution and size. As a method to overcome the effect of these different characteristics in the segmentation, a dual path segmentation framework is proposed in this study. This framework consists of two deep learning models each designed to address the unique features of breast masses and calcifications. The following section provides details on the proposed segmentation approach.

1) *Dual Path Segmentation Framework*: The complete segmentation process consists of a dual path approach as shown in Fig. 2. Initially, the standardized mammogram is sent through the enhancement pipelines explained in the preprocessing section above. Then the mass enhanced mammograms and the calcification enhanced mammograms are used as input for the segmentation process along with the segmentation masks obtained from the annotation files in the dataset.

In the segmentation of breast masses, the preprocessed mammogram images and the segmentation masks used for model training were augmented by applying horizontal flip, vertical flip, zooming and adding Gaussian noise randomly to the samples. Afterwards, the obtained images and corresponding binary segmentation masks

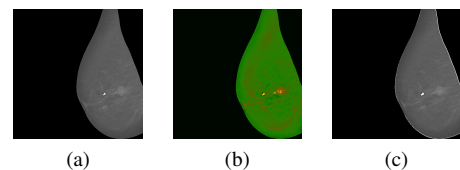


Fig. 1. Comparison of (a) original image with (b) mass enhanced image (c) calcification enhanced image.

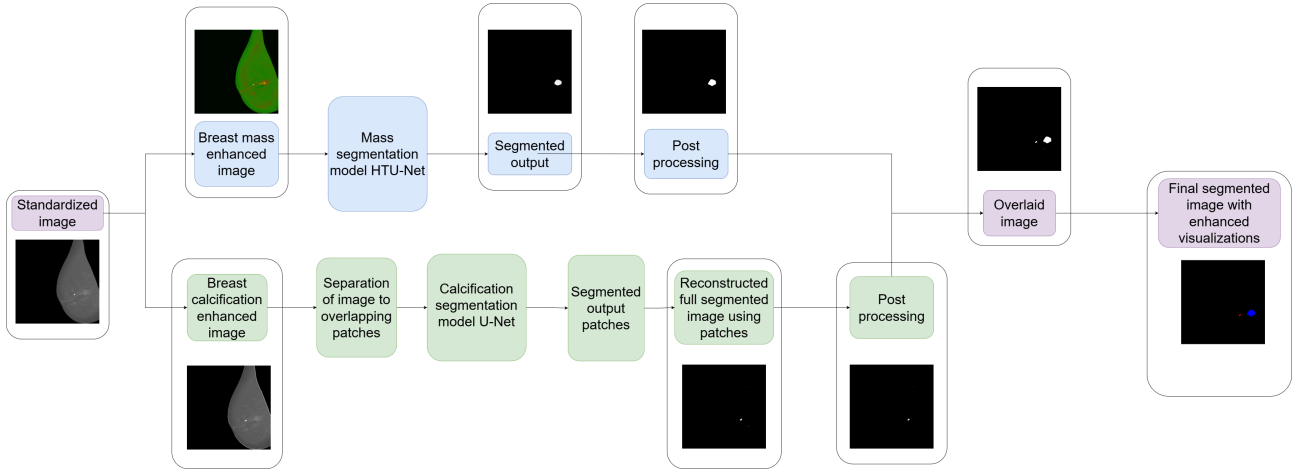


Fig. 2. Flowchart for the proposed segmentation framework including the dual path segmentation for breast masses and calcifications in mammograms

which contain the breast masses are used as input to the segmentation model.

In the segmentation of breast calcifications, the preprocessed images and the corresponding binary segmentation masks including the breast calcifications are separated into overlapping patches and used as input to the segmentation model. The patch based approach consists of extraction of overlapping patches from the image and its corresponding mask after normalization. The details of the patches required for the image reconstruction were stored during the creation of patches. A patch size of 64×64 pixels and a stride of 32 pixels is used and each image with its corresponding segmentation masks are separated into similar sized patches. An experimental threshold value was used to define the patch as positive by considering if the number of positive pixels in each patch exceeds the threshold value. The patches of the image and the patches of the corresponding binary masks with calcifications are used as input to the model.

Afterwards the patches obtained from segmentation are used to reconstruct the complete image. In initializing the reconstruction, an empty image of target size is defined. A counting array tracks the number of times each pixel is updated. The reconstruction process uses a weighted patch merging approach using cosine based- Hanning window [27] technique. The use of Hanning window in this approach served by providing smooth transitions, assisted in handling overlaps and reducing boundary effects. Finally, the reconstructed image is obtained and pixels are normalized.

The outputs obtained by the breast mass segmentation model and the reconstructed images after segmentation of breast calcifications were further smoothed and enhanced by post processing techniques. Then the predicted segmented images obtained for masses and calcifications separately, are overlaid to obtain a final segmented image with both mass and calcification regions.

The following section outlines the details of the architecture of the model selected for segmentation of breast masses and calcifications. Five segmentation architectures were evaluated and compared for mass and calcification segmentation namely, U-Net [28], Hybrid Transformer U-Net (HTU-Net) [29], modified HTU-Net, Attention U-Net

[30] and Attention Recurrent Residual U-Net (AR2U-Net) [31]. Detailed comparisons of performances of these models are included in the results and discussion section.

2) *Model for Segmentation of Breast Masses:* According to the performance of the considered models, the modified HTU-Net was selected as the segmentation model for breast masses. The architecture of the proposed HTU-Net model is as follows. The model has an encoder decoder structure similar to the original U-Net. The encoder consists of a series of convolutional and max pooling layers. The model was improved by introducing a transformer block enhanced with a Spatial-Channel Enhanced Self Attention (SCESA) mechanism in combination with a Multi Head Attention (MHA) at the bottleneck by modifying the location and the type of transformer modules in the original HTU-Net model. The combination of Depthwise Conv2D and MHA modules inside SCESA block is used to extract important features from both local and global aspects of the input. Also, the proposed model uses Conv2D operations as a proxy for positional encoding in contrast to the original HTU-Net which used fixed positional encodings. The proposed modified HTU-Net model does not use the traditional transformer based patch embedding mechanism. It uses the 2D convolutional feature maps that are able to preserve spatial coherence instead of flattening or patch division.

The proposed modified HTU-Net introduces several improvements over the original HTU-Net by reducing the computational overhead, preserving the spatial structure, simplifying positional encoding, and customized attention mechanism. According to the results of the model for breast mass segmentation, it can be observed that the modified HTU-Net is efficient, lightweight and more appropriate for mammogram segmentation for extracting breast masses.

The model accepts the preprocessed mammogram images of size $1024 \times 1024 \times 3$ and the final convolutional layer with sigmoid activation outputs a binary mask indicating the mass regions.

3) *Model for Segmentation of Breast Calcifications:* When considering the outputs obtained for calcification segmentation, and the values of performance evaluation

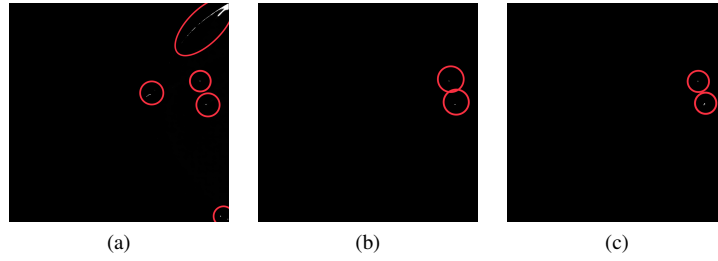


Fig. 3. Comparison of the (a) reconstructed image, (b) post-processed image, and (c) ground truth mask. The red colour markings are used for better visualization.

metrics U-Net was observed to be the best performing model. The encoder of the U-Net model consists of three convolutional blocks. At the bottleneck two convolutional layers with 512 filters each are applied. The decoder of this model mirrors the encoder with three upsampling blocks.

The model accepts the greyscale image patches of size $64 \times 64 \times 1$ and the final layer with sigmoid activation function outputs the patches of the binary mask with calcifications.

4) *Post Processing and Output Fusion*: The post processing steps followed are explained in detail in the following section. For the output obtained by the mass segmentation, thresholding is used as the initial post processing step. This converts the probability maps obtained by neural networks to binary masks by deciding whether the pixel value exceeds the threshold or not. Then the small isolated regions that are likely to be noise or false positives are eliminated. Morphological smoothing is done by the closing operation and it is used to smoothen the boundaries and fill small gaps. Finally the largest connected region in the mask is kept by removing all the other small disconnected regions. The reconstructed images obtained after segmentation of calcifications are post processed by a different approach. The initial pre-processed image is processed with morphological closing and the breast boundary is removed from this image. Both the obtained image and predicted segmentation mask is normalized and these images are multiplied with each other to obtain a better segmentation output. Fig. 3 shows the comparison of reconstructed image from segmented patches, post processed image and the ground truth mask for calcifications

The post processed outputs through the two segmentation paths are overlaid on each other for fusion of two outputs. For an enhanced and clear visualization, the two types of abnormalities are shown in two different colours.

5) *Evaluation Metrics for Segmentation Models*: The Dice Similarity Coefficient (DSC), Jaccard Index, Average Surface Distance (ASD), and precision were used as the evaluation metrics were used to evaluate the performance of the segmentation models.

IV. RESULTS AND DISCUSSION

The results and discussion section compares the performance of the four different models used for segmentation of masses and calcifications.

According to the results shown in Table I, the HTU-Net has outperformed all the other models for breast mass segmentation. It has obtained the highest DSC, Jaccard

TABLE I
COMPARISON OF PERFORMANCE OF MODELS USED FOR SEGMENTATION OF BREAST MASSES

Model	DSC	Jaccard Index	ASD	Precision
U-Net	0.5650	0.3854	0.0094	0.5563
AR2U-Net	0.4660	0.2935	0.0090	0.4580
Attention U-Net	0.3668	0.2775	0.0120	0.4344
HTU-Net	0.5148	0.3644	0.0091	0.5556
Modified HTU-Net	0.6064	0.4273	0.0084	0.5988

TABLE II
COMPARISON OF PERFORMANCE OF MODELS USED FOR SEGMENTATION OF BREAST CALCIFICATIONS

Model	DSC	Jaccard Index	ASD	Precision
AR2U-Net	0.7411	0.6544	0.0221	0.8944
Attention U-Net	0.7241	0.6449	0.0232	0.8670
HTU-Net	0.7310	0.5785	0.0242	0.7692
Modified HTU-Net	0.8099	0.6806	0.0204	0.9007
U-Net	0.8022	0.6838	0.0203	0.8958

index and precision with the lowest ASD among all the candidate models. Also, the DSC of mass segmentation has a relatively low value than the DSC for calcification segmentation as there were small differences in the segmented mass shape and the shape of the mass in the ground truth. However the breast masses were accurately localized by the segmentation model. Table IV shows the comparison between post processed segmentation output of the proposed method for masses and the ground truth masks for masses.

Table II shows the comparison of model performance for segmentation of breast calcifications using the patch based approach. According to the results, the U-Net and the HTU-Net performs better than the other two models. However, the DSC of the selected U-Net model for calcification segmentation is less than that of the HTU-Net. The U-Net model was selected by considering both the performance evaluation metrics values and the structural similarity of the segmented output to the original annotation. Table III shows the comparison between the segmented images from U-Net, HTU-Net and the original segmentation mask. Calcifications in these images are shown as tiny white dots against the black background. When comparing these images visually, it can be observed that the outputs obtained by HTU-Net has captured unwanted noise even if the evaluation metric results are relatively better than U-Net. Accordingly, U-Net was considered the most accurate segmentation model for calcifications among the four candidate models.

TABLE III
VISUAL COMPARISON OF SEGMENTATION RESULTS OF HTU-NET AND U-NET WITH THE GROUND TRUTH FOR CALCIFICATION SEGMENTATION

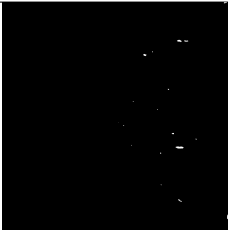
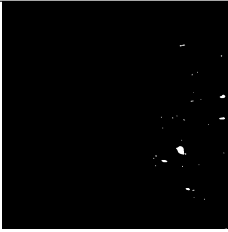
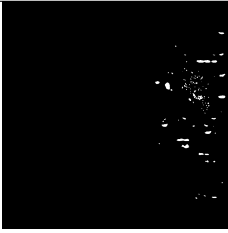
HTU-Net Output (After post processing)	U-Net Output (After post processing)	Ground Truth Mask
		
		
		

TABLE IV
VISUAL COMPARISON OF BREAST MASS SEGMENTATION RESULTS OF HTU-NET AND GROUND TRUTH MASK

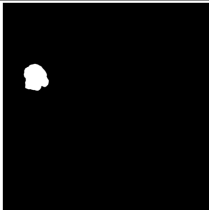
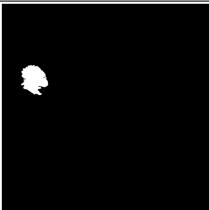
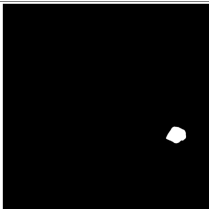
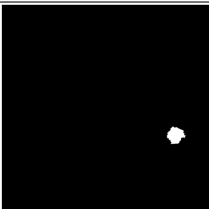
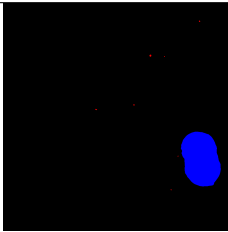
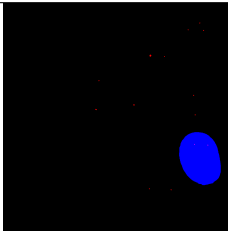
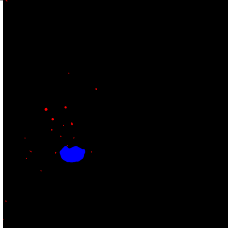
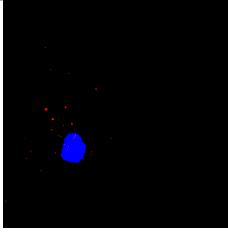
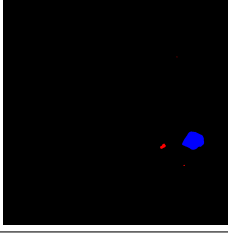
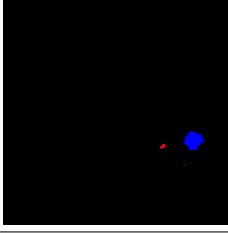
Output of mass segmentation by proposed method (Post processed)	Ground Truth
	
	
	

TABLE V
VISUAL COMPARISON OF FINAL SEGMENTATION RESULTS OF PROPOSED METHOD AND GROUND TRUTH MASK

Final segmentation by proposed method	Ground Truth
	
	
	

The final segmented image was obtained by overlaying the post processed outputs from the separate segmentation paths. To enhance the visualization of the masses and calcifications two different colours were used. Table V shows the comparison between the final segmentation masks

predicted and processed, and the original segmentation masks. It can be observed that the proposed approach has the potential to extract the ROIs and accurately segment masses and calcifications in mammograms for

the accurate diagnosis of breast cancer.

V. CONCLUSION

The study focused on designing a unified approach for breast mass and calcification segmentation in mammograms to assist in accurate diagnosis of breast cancer. The proposed approach is a dual path segmentation framework with two separate workflows tailored for segmentation of each specific abnormality. Four DL based models were evaluated for the segmentation task. According to the quantitative and qualitative results, the modified HTU-Net achieved the highest performance for segmentation of breast masses and U-Net obtained the best performance for segmentation of breast calcifications. The calcification segmentation was performed through a patch based approach. The final stage of the framework includes the post processed output fusion from the two segmentation pathways. The results obtained through the comparison of predicted segmentation masks with the ground truth proves that the proposed approach captures the fine structural details, and has a superior performance in segmentation of both breast masses and calcifications. Future work of this study will explore the possibility of using the segmentation results for accurate classification of the abnormality according to the Breast Imaging-Reporting and Data System (BI-RADS) categories. In conclusion, the proposed dual path segmentation framework has the potential of accurately segmenting breast masses and calcifications to assist radiologists in accurate and efficient diagnosis of breast cancer.

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